Vaishnav Dhruv

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.18
Sri prakash synergy school	Class XII (CBSE)	2022	91.20%
Sri prakash synergy school	Class X (CBSE)	2020	93.40%

EXPERIENCE

ServiceNow - Machine Learning Intern, Agentic AI

- Built an end-to-end evaluation workflow for autonomous-agent executions, replacing manual spreadsheet review with repeatable LLM-as-a-Judge scoring across RCA accuracy, solution quality, citation fidelity, and customer-question handling; added semantic comparison, side-by-side AI vs expert analysis, and aggregate reporting.
- Developed debugging and observability tooling for hierarchical multi-agent workflows by reconstructing agent-tool relationships from platform metadata and tracing retrieved-document propagation to localize retrieval, context-loss, and reasoning failures.
- Built a configuration validator for customizable AI workers, recursively checking JSON-defined tables, fields, agent settings, and required platform dependencies against ServiceNow metadata.
- Contributed agent-workflow reliability fixes, including normalizing inconsistent structured LLM outputs into the representation expected by downstream backend processing.

RESEARCH EXPERIENCE

FIGScore: A Format-Invariant Geospatial Similarity Score for Evaluating How Small Vision-Language Models Read Precipitation Maps - *Submitted to NeurIPS 2026 Workshop on Tackling Climate Change with ML*

- Ran experiments evaluating five vision-language models (256M-7B parameters) on NASA GPM IMERG precipitation maps from seven hurricane events using 77 image-question pairs and 385 generated responses.
- Contributed to FIGScore, a format-invariant geospatial metric that normalizes and scores factual agreement across event names, locations, coordinates, date ranges, precipitation amounts, units, and color-scale values.
- Performed metric analysis comparing FIGScore against BLEU, chrF, ROUGE, METEOR, embedding cosine similarity, and BERTScore; the study reported a 14.0x correct-to-wrong score ratio and 0.0 format drift.

PROJECTS

RepoLens - Grounded Technical Documentation Copilot

- Built a citation-grounded RAG system for **GitHub repositories and technical documents**, preserving exact repository paths, source-code line ranges, and PDF page references in generated answers.
- Implemented **hybrid BM25 + dense retrieval**, persistent embedding collections, optional cross-encoder reranking, and offline/local generation with graceful fallback when an LLM is unavailable.
- Developed a **FastAPI service, Streamlit client, Docker Compose deployment, CI pipeline, and retrieval evaluation suite** reporting Recall@K, MRR, citation rate, and latency metrics.

Distributed Job Queue & Scheduler

- Built a distributed task scheduler with priority/delayed jobs, idempotent APIs, transactional outbox delivery, retries, dead-letter handling, and Prometheus observability.
- Implemented **fault-tolerant execution** using **atomic PostgreSQL claims, renewable leases, and fencing tokens**, with automatic recovery from worker crashes.
- Stress-tested **10K jobs across 8 workers**, reaching **285.4 completions/sec with zero lost jobs or duplicate completions**.

Self-Learning Coding Agent Harness

- Built an **iterative coding agent** that generates solutions, executes untrusted code in **isolated Docker sandboxes**, and evaluates attempts using deterministic tests, property-based checks, and structured failure feedback.
- Implemented reflection and **versioned lesson memory** that converts failures into reusable strategies, retrieves

relevant lessons for new tasks, and supports rejection, superseding, and rollback of harmful memories.
- Designed **validation-gated learning** with separate training, lesson-validation, and final-validation task sets, retaining lessons only when they demonstrate improvement on held-out tasks.
- Added reproducible experiment tracking for model and prompt versions, retrieved memories, test outcomes, latency, token usage, and **before-vs-after learning performance**.

SKILLS

Programming: Python, C/C++

ML & Data: NumPy, Pandas, SciPy, Matplotlib, DeepXDE, KMeans, KNN

AI & Engineering: LLM evaluation, multi-agent systems, RAG, LangChain, Streamlit, FAISS, Chroma, Hugging Face Transformers, Ollama, Git/GitHub, LaTeX

POSITIONS OF RESPONSIBILITY

Multimedia Head, Diesta'25 - Interdepartmental Fest, IITH

Led the multimedia team in developing promotional content, event coverage, and post-fest highlights.

Multimedia Coordinator, Ek Bharat Shreshtha Bharat (EBSB)

Documented cultural and academic events throughout the EBSB tenure via photography and videography.

Core Member, Infocus Photography Club, IITH

Participated in workshops and campus photo walks to enhance photography and editing skills.